Kim 2018 TNBC — analysis summary

RUN 1 OF 1

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: same-sample neighbour fraction 0.99→0.40; iLISI 3.89 of 9 samples; continuity confirmed off the independent PCA-harmony graph (1 connected component), 0 islands.

Gate 2 — module geography → GO (non-branching)

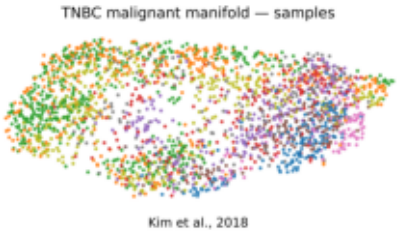
Single axis: pSom-high origin → Gast/Crest-high terminus. Organized Gast↔pSom cross-repression (smoothed r=−0.84); Crest a terminal gradient. FLAG: Crest enriches the Gast pole (inversion) — passes anyway.

Gate 3 — cross-repression → NA

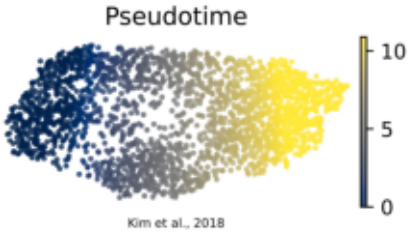
No bifurcation (curve: 2 tips, 0 forks) — branching-only Gate 3 not applicable.

STEP FIGURES

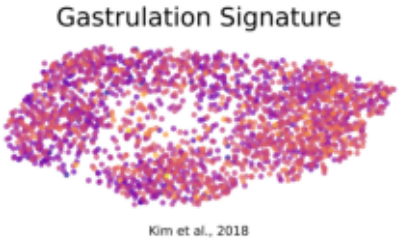
Manifold — samples



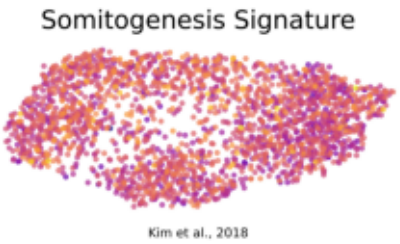
Pseudotime



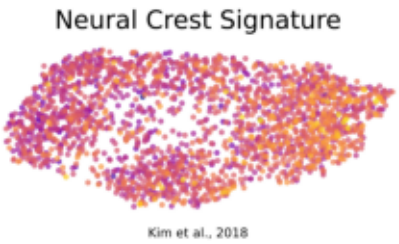
Gastrulation



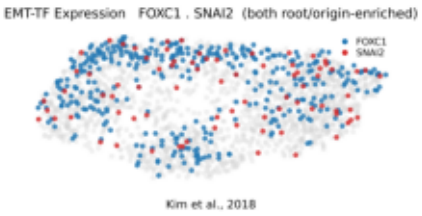
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Triple-negative breast cancer (Kim et al., 2018), Nanogrid — 2,263 malignant cells across 9 samples (9/16 kept at ≥100-cell floor). Non-branching Linear-pSom/Crest.

Verdict.
The embryonic-program pattern reproduces as a non-branching linear axis: a single continuous, sample-integrated manifold from a pSom-high developmental origin to a Gast/Crest-high terminus, with organized Gast↔pSom cross-repression and Crest as a terminal gradient enriched on the Gast pole (inversion flag). No bifurcation → Gate 3 NA. EMT-TF expression (FOXC1, SNAI2) concentrates at the root/origin end; no EMT-TF marks the terminus in this dataset.

Parameter choices

Min-cell threshold	≥100 (floor coincides with real 52→100 inflection) → 9/16, 2,263 cells	**Embedding**	HVG=536; PCA50; Harmony θ=10 (TNBC patient batch); n_pcs=20, spread=1.3
Trajectory	scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.98)	**Root**	pSom-high/Crest-low developmental origin (Crest is late)
Geometry	Linear-pSom/Crest (raw Gast −0.009, pSom +0.040, Crest −0.085)	**EMT-TFs**	FOXC1 (65%, ρ=−0.41) + SNAI2 (ρ=−0.23); BOTH root/origin-enriched, none terminal